

KEMI AJOS, Finland

WMO: 028620

Lat: 65.673N

Lon: 24.515E

Elev: 10

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.0	-8.3	-18.3	2.0	-12.3	-12.8	2.7	-8.2	37.4	31.7	34.5	31.2	8.9	120	0.817

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	74.6	64.8	71.2	63.7	68.4	61.8	67.3	72.8	64.8	69.7	62.7	67.4	9.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	92.5	70.5	62.6	84.9	68.0	60.6	79.0	65.3	31.7	73.2	29.9	69.6	28.3	67.6	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.9	30.0	26.6	-16.8	79.0	6.3	4.6	-21.3	82.3	-25.0	85.0	-28.5	87.6	-33.0	90.9		
(5)			-17.2	68.4	6.2	3.1	-21.6	70.6	-25.2	72.4	-28.7	74.2	-33.2	76.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	16.0	15.1	21.1	31.7	44.1	54.7	61.9	59.3	51.1	39.3	30.3	24.3	(6)
(7)		DBStd	17.99	11.77	12.79	9.34	5.63	7.59	5.56	5.33	5.05	5.32	6.63	8.27	10.16	(7)
(8)		HDD50	5479	1054	978	896	548	217	17	0	2	50	332	590	795	(8)
(9)		HDD65	10064	1519	1398	1361	998	649	312	127	188	416	796	1040	1260	(9)
(10)		CDD50	936	0	0	0	0	34	158	369	290	84	1	0	0	(10)
(11)		CDD65	46	0	0	0	0	1	4	31	10	0	0	0	0	(11)
(12)		CDH74	127	0	0	0	0	5	13	95	14	0	0	0	0	(12)
(13)		CDH80	13	0	0	0	0	0	1	12	0	0	0	0	0	(13)
(14)	Wind	WSAvg	13.0	13.3	13.3	13.3	11.9	11.8	12.0	11.1	12.2	14.2	14.6	14.5	14.3	(14)
(15)	Precipitation	PrecAvg	23.4	2.0	1.6	1.3	1.2	1.5	1.8	2.7	2.4	2.1	2.2	2.4	2.0	(15)
(16)		PrecMax	32.8	3.6	3.3	2.4	3.1	2.7	3.2	5.4	6.2	4.7	4.2	4.4	4.6	(16)
(17)		PrecMin	17.4	0.4	0.1	0.4	0.3	0.4	0.3	0.3	0.5	0.6	0.9	0.7	0.6	(17)
(18)		PrecStd	4.7	0.9	0.9	0.6	0.7	0.6	0.9	1.3	1.5	1.1	0.9	1.0	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.4	36.6	38.5	51.2	72.5	75.5	80.9	75.8	64.6	52.7	45.2	39.2	(19)
(20)			MCWB	34.2	34.1	34.3	42.5	58.4	62.2	68.0	66.0	57.9	50.1	43.6	37.9	(20)
(21)		2%	DB	34.0	34.7	36.0	44.9	64.9	69.5	76.4	71.4	61.1	51.0	43.9	37.2	(21)
(22)			MCWB	33.3	33.3	33.5	38.6	54.4	59.2	66.2	64.6	56.6	49.0	42.9	35.8	(22)
(23)		5%	DB	33.0	33.2	34.6	40.8	59.7	65.6	72.9	68.7	59.4	49.7	41.6	36.1	(23)
(24)			MCWB	32.4	32.4	32.8	36.3	51.4	57.4	65.1	62.3	56.2	47.7	40.3	35.0	(24)
(25)		10%	DB	31.9	31.8	33.3	38.2	54.4	62.6	70.1	66.3	58.3	48.2	39.8	35.0	(25)
(26)			MCWB	31.4	31.2	32.0	34.9	48.1	56.2	63.8	61.1	55.3	46.2	38.7	34.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.6	34.9	35.1	44.2	60.4	65.0	70.5	69.0	60.4	51.1	44.4	38.3	(27)
(28)			MCDWB	35.1	36.4	36.8	49.0	69.5	71.5	75.7	73.5	63.1	52.0	44.6	38.9	(28)
(29)		2%	WB	33.6	33.6	34.0	39.1	55.6	61.4	68.6	65.9	58.0	49.6	43.1	36.2	(29)
(30)			MCDWB	34.0	34.4	35.3	44.1	64.2	67.3	74.1	70.4	59.8	50.8	43.9	36.7	(30)
(31)		5%	WB	32.6	32.5	33.2	36.9	51.8	59.1	66.4	63.5	56.9	48.1	40.4	35.2	(31)
(32)			MCDWB	33.0	33.2	34.2	39.6	57.9	63.9	72.1	67.2	59.0	49.4	41.3	35.8	(32)
(33)		10%	WB	31.5	31.2	32.2	35.5	49.1	57.2	64.1	61.7	55.7	46.5	38.9	34.2	(33)
(34)			MCDWB	31.9	31.9	33.2	37.6	53.9	61.3	69.2	65.2	58.0	48.2	39.8	34.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.2	11.8	13.1	10.8	10.3	9.0	9.3	8.3	7.1	6.3	6.9	8.5	(35)
(36)			MCDBR	8.5	8.0	10.1	13.5	17.2	13.5	13.2	11.5	7.9	5.0	5.7	5.7	(36)
(37)			MCWBR	8.3	7.4	8.6	9.0	9.5	7.7	7.6	6.8	5.8	4.9	5.7	5.7	(37)
(38)		5% WB	MCDWB	8.7	8.4	9.3	11.1	15.7	11.3	11.0	8.6	6.5	4.9	5.7	5.5	(38)
(39)			MCWBR	8.6	8.0	8.2	7.9	9.8	7.5	7.4	6.2	5.2	4.8	5.7	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.207	0.273	0.303	0.339	0.346	0.349	0.367	0.351	0.328	0.304	0.343	0.355	(40)	
(41)		taud	1.943	2.088	2.139	2.192	2.413	2.482	2.454	2.486	2.529	2.347	2.564	2.603	(41)	
(42)		Ebn at Noon	80	184	239	260	272	274	264	258	239	179	76	25	(42)	
(43)		Edh at Noon	6	18	28	35	31	30	30	26	20	15	6	3	(43)	
(44)	All-Sky Solar Radiation	RadAvg	26	180	648	1201	1567	1757	1633	1192	636	228	44	5	(44)	
(45)		RadStd	2	36	66	95	141	133	143	109	67	38	7	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page